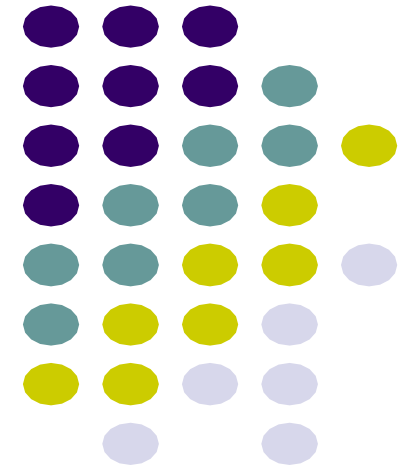


PROJECT COMMUNICATIONS MANAGEMENT





What is SW project communications management?

- A subset of project management that includes the processes required to ensure *timely* and *appropriate* generation, collection, distribution, storage, and ultimate character of project information.



Importance of Good Communications

- The greatest **threat** to many projects is a failure to communicate
- Our culture does not describe SW professionals as being good communicators
- Research shows that SW professionals must be able to communicate effectively to succeed in their positions
- Strong verbal skill is a key factor in career advancement for SW professionals



Project Communications Management Processes

1. **Communications planning:** determining the information and communication needs of the stakeholders
2. **Information distribution:** making needed information available in a timely manner
3. **Performance reporting:** collecting and distributing performance information
4. **Administrative closure:** generating, gathering, and disseminating information to formalize phase or project completion



Communications Planning

- Every project should include **some type of communications management plan**, a document that **guides project communications**
- **Creating a stakeholder analysis** for project communications also aids in communications planning



Communications Management Plan Contents

- A description of a collection and filing structure for gathering and storing various types of information
- A distribution structure describing **what** information goes to **whom**, **when**, and **how**
- A **format** for communicating key project information
- A **project schedule** for producing the information



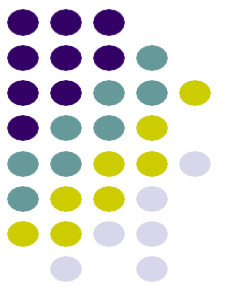
Sample Stakeholder Analysis for Project Communications

Stakeholders	Document Name	Document Format	Contact Person	Due
Customer Management	Monthly Status Report	Hard copy	Gail Feldman, Tony Silva	First of month
Customer Business Staff	Monthly Status Report	Hard copy	Julie Grant, Jeff Martin	First of month
Customer Technical Staff	Monthly Status Report	E-mail	Evan Dodge, Nancy Michaels	First of month
Internal Management	Monthly Status Report	Hard copy	Bob Thomson	First of month
Internal Business and Technical Staff	Monthly Status Report	Intranet	Angie Liu	First of month
Training Subcontractor	Training Plan	Hard Copy	Jonathan Kraus	11/1/1999
Software Subcontractor	Software Implementation Plan	E-mail	Barbara Gates	6/1/2000

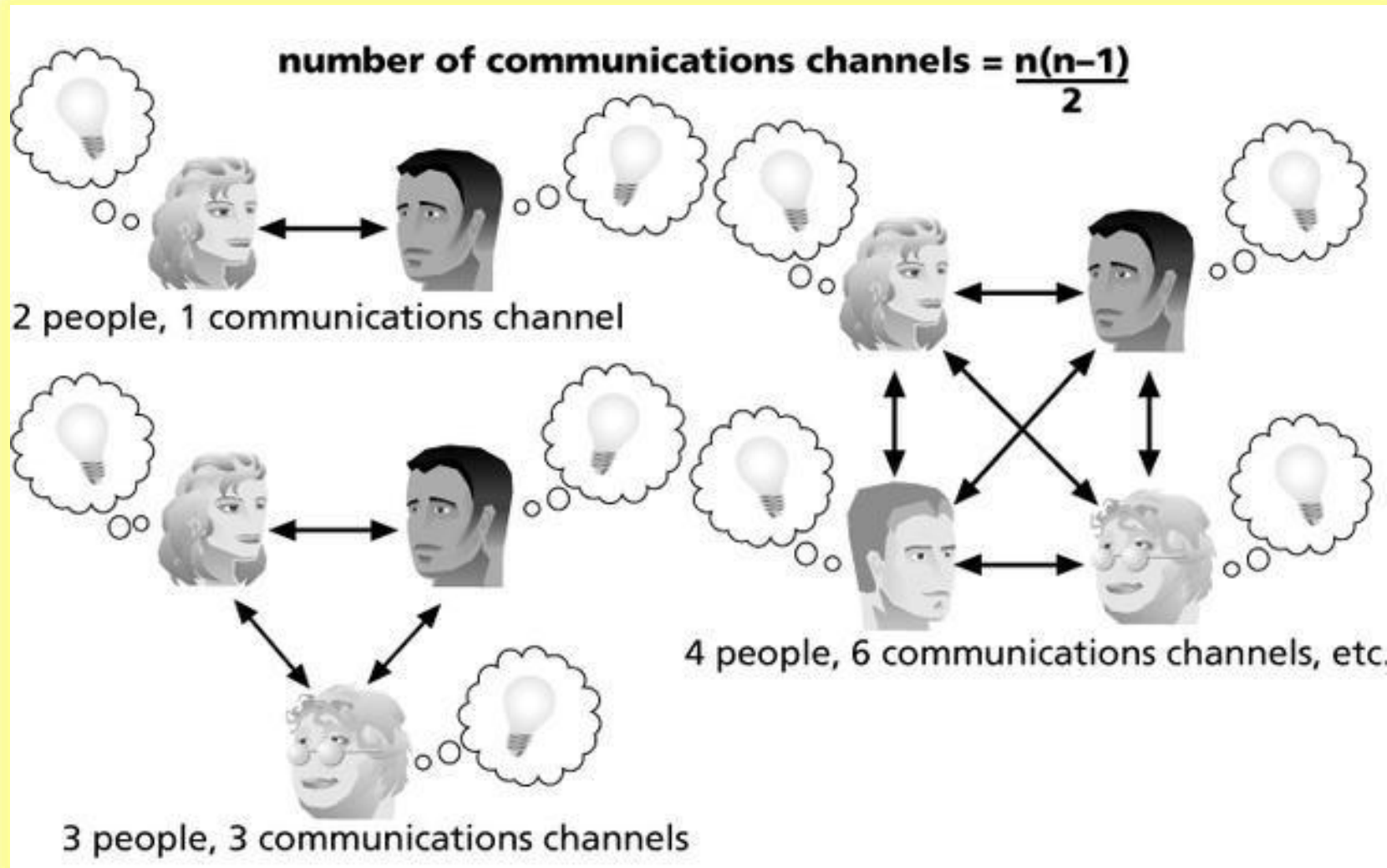


Information Distribution

- Getting the *right information* to the *right people* at the *right time* and in a *useful format* is just as important as developing the information in the first place
- Important considerations include
 - using technology to enhance information distribution
 - formal and informal methods for distributing information



THE IMPACT OF THE NUMBER OF PEOPLE ON COMMUNICATIONS CHANNELS





Performance Reporting

- Performance reporting keeps stakeholders informed about how resources are being used to achieve project objectives.
- For example
 - **Status reports** describe where the project stands at a specific point in time
 - **Progress reports** describe what the project team has accomplished during a certain period of time
 - **Project forecasting** predicts future project status and progress based on past information and trends
 - **Status review meetings** often include performance reporting



Administrative Closure

- It is concerned with generating, gathering, and disseminating information to formalize phase or project completion
- A project or phase of a project requires closure
- Administrative closure produces
 - Project archives
 - Formal acceptance
 - Lessons learned



Conflict Can Be Good

- Conflict often produces important results, such as new ideas, better alternatives, and motivation to work harder and more collaboratively
- **Groupthink** can develop if there are no conflicting viewpoints
- Research by Karen Jehn suggests that **task-related conflict** often improves team performance, but **emotional conflict** often depresses team performance



Conflict Handling Modes, in Preference Order

1. **Confrontation** or **problem-solving**: directly face a conflict(Discuss, solve)
2. **Compromise**: use a give-and-take approach(*Give up, halfway*)
3. **Smoothing**: de-emphasize areas of difference and emphasize areas of agreement(*Calm, agree*)
4. **Forcing**: the win-lose approach(*Force, command*)
5. **Withdrawal**: retreat or withdraw from an actual or potential disagreement(*Avoid, ignore*)



Running Effective Meetings

- Define the purpose and intended outcome of the meeting
- Determine who should attend the meeting
- Provide an agenda to participants before the meeting
- Prepare handouts, visual aids, and make logistical arrangements ahead of time
- Run the meeting professionally
- Build relationships



DEVELOPING A COMMUNICATIONS INFRASTRUCTURE

- A communications infrastructure is a **set of tools, techniques, and principles** that provide a **foundation for the effective transfer of information**
 - Tools include e-mail, project management software, groupware, fax machines, telephones, teleconferencing systems, document management systems, and word processors
 - Techniques include **reporting guidelines and templates**, meeting ground rules and procedures, decision-making processes, problem-solving approaches, and conflict resolution and negotiation techniques
 - Principles include using open **dialogue and an agreed-upon work** ethics



Management



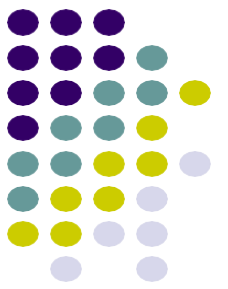
What is Risk?

- A dictionary definition of risk is “the possibility of loss or injury”
- “A risk is a combination of **constraint** and **uncertainty**” by Larry Krantz
- Project risk management involves **understanding potential** problems that might occur on the project and **how they might hinder project success**
- Risk management is like a **form of insurance**; it is an investment.

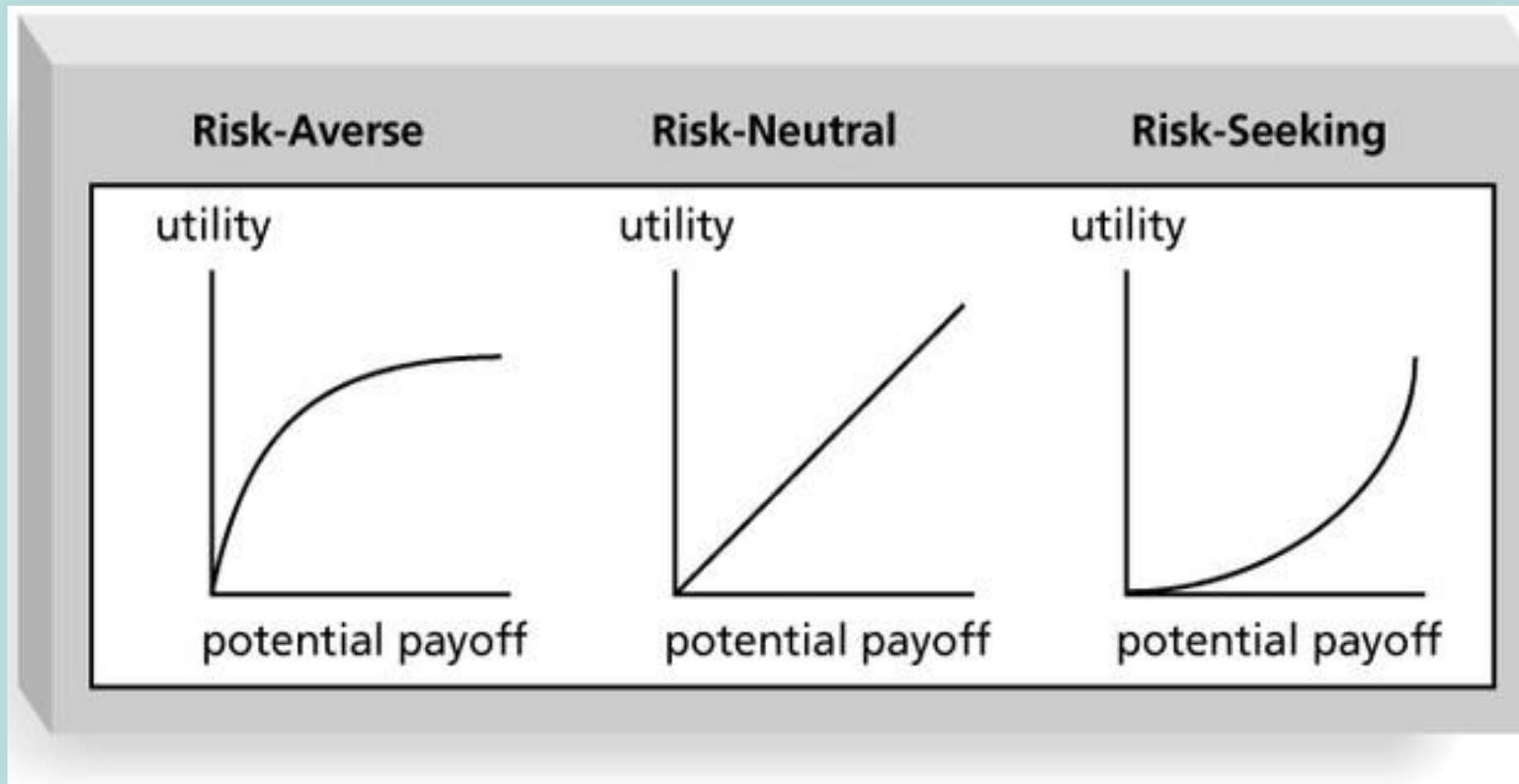
Risk Utility



- Risk utility or risk tolerance is the amount of satisfaction or pleasure received from a potential payoff
 - Utility rises at a decreasing rate for a person who is **risk-averse**
 - Those who are **risk-seeking/taking** have a higher tolerance for risk, and their satisfaction increases when more payoff is at stake
 - The **risk-neutral** approach achieves a balance between risk and payoff



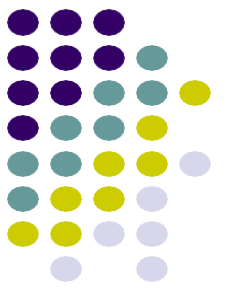
Risk Utility Function and Risk Preference





WHAT IS PROJECT RISK MANAGEMENT?

- Risk management is the systematic process of *identifying, analyzing, and responding* to project risk.
- It includes **maximizing the probability and consequences of positive events** and **minimizing** the probability and consequences of **adverse events** to project objectives.



Project risk management processes

The goal of project risk management is to **minimize potential risks** while maximizing **potential opportunities**. Major processes include

- 1. Risk management planning:** **deciding** how to approach and plan the risk management activities for the project
- 2. Risk identification:** **determining** which risks are likely to affect a project and documenting their characteristics
- 3. Qualitative risk analysis:** **characterizing** and **analyzing** risks and prioritizing their effects on project objectives

...processes

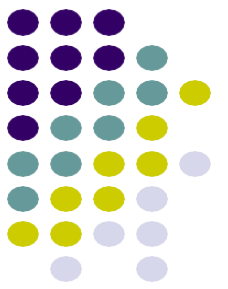


4. **Quantitative risk analysis:** measuring the probability and consequences of risks
5. **Risk response planning:** taking steps to enhance opportunities and reduce threats to meeting project objectives
6. **Risk monitoring and control:** monitoring known risks, identifying new risks, reducing risks, and evaluating the effectiveness of risk reduction

Risk Management Planning

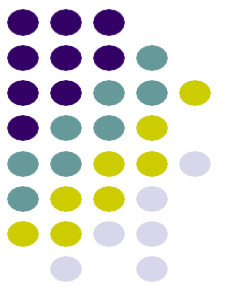


- The project team should review project documents and understand the organization's and the sponsor's approach to risk
- The level of detail will vary with the needs of the project



Questions Addressed in a Risk Management Plan

- **Why** is it important to take/not take this risk in relation to the project objectives?
- **What** is the specific risk, and what are the risk mitigation deliverables?
- **How** is the risk going to be mitigated? (What risk mitigation approach is to be used?)
- **Who** are the individuals responsible for implementing the risk management plan?
- **When** will the milestones associated with the mitigation approach occur?
- **How much** is required in terms of resources to mitigate risk?



Risk Identification

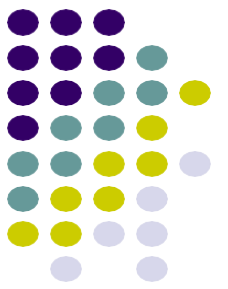
- Risk identification is the process of understanding what potential unsatisfactory outcomes are associated with a particular project
- Several risk identification tools and techniques include
 - Interviewing
 - Brainstorming – group discussion
 - Delphi technique – panel of experts
 - SWOT analysis

Read more on these
tools and techniques

Potential Risk Conditions Associated With Each Knowledge Area



Knowledge Area	Risk Conditions
Integration	Inadequate planning; poor resource allocation; poor integration management; lack of post-project review
Scope	Poor definition of scope or work packages; incomplete definition of quality requirements; inadequate scope control
Time	Errors in estimating time or resource availability; poor allocation and management of float; early release of competitive products
Cost	Estimating errors; inadequate productivity, cost, change, or contingency control; poor maintenance, security, purchasing, etc.
Quality	Poor attitude toward quality; substandard design/materials/workmanship; inadequate quality assurance program
Human Resources	Poor conflict management; poor project organization and definition of responsibilities; absence of leadership
Communications	Carelessness in planning or communicating; lack of consultation with key stakeholders
Risk	Ignoring risk; unclear assignment of risk; poor insurance management
Procurement	Unenforceable conditions or contract clauses; adversarial relations



Qualitative Risk Analysis

- Assess the **probability** and **impact of identified** risks to determine their **magnitude** and **priority**
- Risk quantification tools and techniques include
 - Probability/Impact matrixes
 - The Top 10 Risk Item Tracking technique
 - Expert judgment



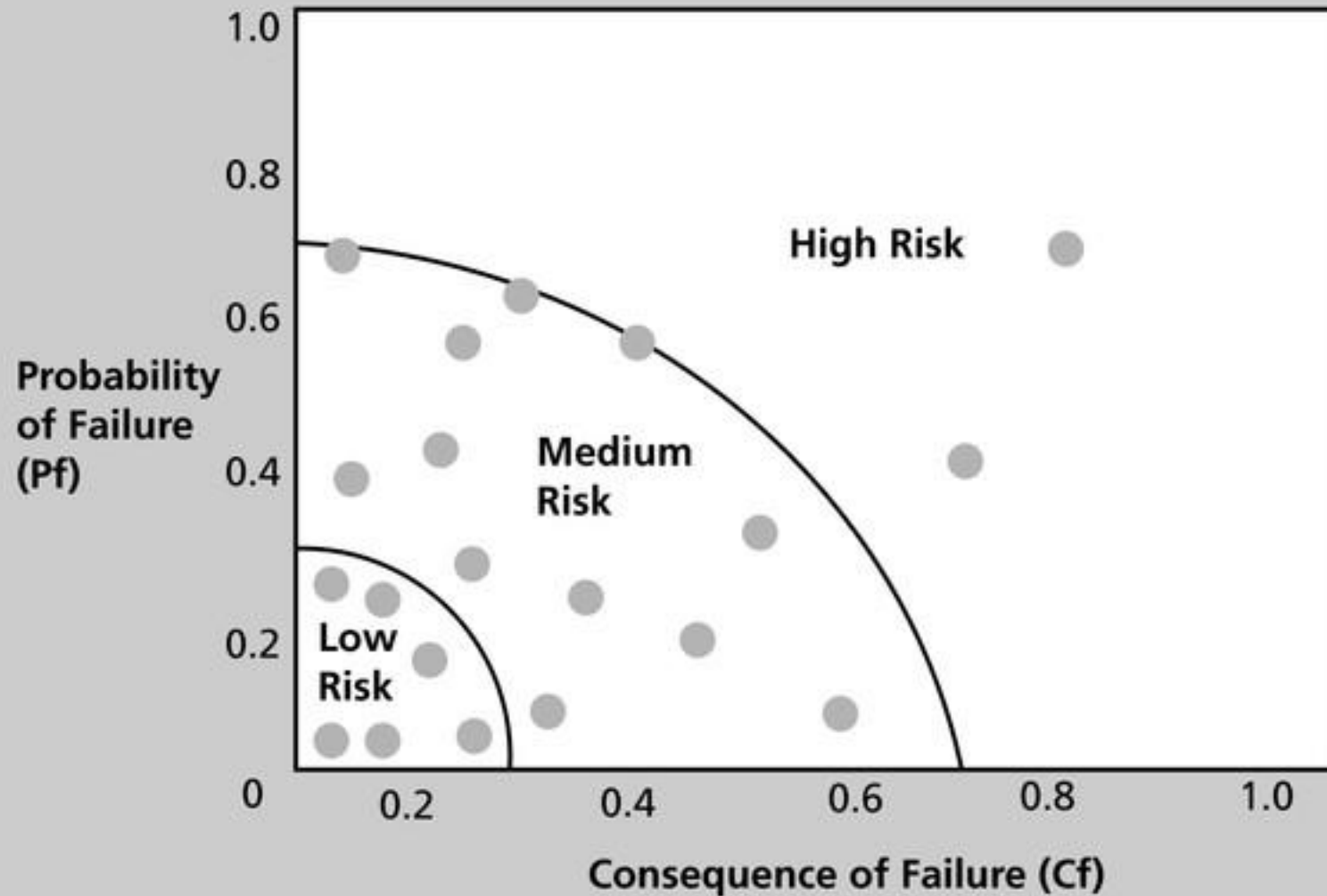
Sample Probability/Impact Matrix for Qualitative Risk Assessment

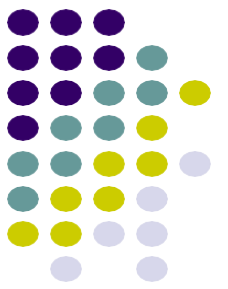


PROBABILITY OF FAILURE (P _F) ATTRIBUTES OF SUGGESTED TECHNOLOGY			
VALUE	MATURITY HARDWARE/SOFTWARE	COMPLEXITY HARDWARE/SOFTWARE	SUPPORT BASE
0.1	Existing	Simple Design	Multiple Programs And Services
0.3	Minor Redesign	Somewhat Complex	Multiple Programs
0.5	Major Change Feasible	Fairly Complex	Several Parallel Programs
0.7	Complex HW Design/ New SW Similar to Existing	Very Complex	At Least One Other Program
0.9	Some Research Completed/ Never Done Before	Extremely Complex	No Additional Programs

CONSEQUENCE OF FAILURE (C _F) ATTRIBUTES OF SUGGESTED TECHNOLOGY				
VALUE	FALLBACK SOLUTIONS	LIFE CYCLE COST (LCC) FACTOR	SCHEDULE FACTOR (INITIAL OPERATIONAL CAPABILITY = IOC)	DOWNTIME (DT) FACTOR
0.1	Several Acceptable Alternatives	Highly Confident Will Reduce LCC	90—100% Confident Will Meet IOC Significantly	Highly Confident Will Reduce DT
0.3	A Few Known Alternatives	Fairly Confident Will Reduce LCC	75—90% Confident Will Meet IOC	Fairly Confident Will Reduce DT Significantly
0.5	Single Acceptable Alternative	LCC Will Not Change Much	50—75% Confident Will Meet IOC	Highly Confident Will Reduce DT Somewhat
0.7	Some Possible Alternatives	Fairly Confident Will Increase LCC	25—50% Confident Will Meet IOC	Fairly Confident Will Reduce DT Somewhat
0.9	No Acceptable Alternatives	Highly Confident Will Increase LCC	0—25% Confident Will Meet IOC	DT May Not Be Reduced Much

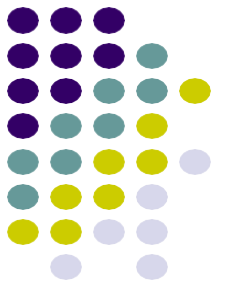
Chart Showing High, Medium, and Low-Risk Technologies





Top 10 Risk Item Tracking

- Top 10 Risk Item Tracking is a tool for maintaining an awareness of risk throughout the life of a project
- Establish a periodic review of the top **10 project risk items**
- List the **current ranking, previous ranking, number of times** the risk appears on the **list over a period of time**, and a summary of progress made in resolving the risk item



Example of Top 10 Risk Item Tracking

	Monthly Ranking			
Risk Item	This Month	Last Month	Number of Months	Risk Resolution Progress
Inadequate planning	1	2	4	Working on revising the entire project plan
Poor definition of scope	2	3	3	Holding meetings with project customer and sponsor to clarify scope
Absence of leadership	3	1	2	Just assigned a new project manager to lead the project after old one quit
Poor cost estimates	4	4	3	Revising cost estimates
Poor time estimates	5	5	3	Revising schedule estimates

Expert Judgment



- Many organizations rely on the in-built feelings and past experience of experts to help identify potential project risks
- Experts can categorize risks as **high**, **medium**, or **low** with or without more sophisticated techniques

Quantitative Risk Analysis



- Often follows qualitative risk analysis, but both can be done together or separately
- Large, complex projects involving leading edge technologies often require extensive quantitative risk analysis
- Main techniques include
 - Decision tree analysis
 - Simulation

Decision Trees and Expected Monetary Value (EMV)



- A decision tree is a **diagramming method** used to help you select the best course of action in situations in **which future outcomes** are uncertain
- EMV is a type of decision tree where you calculate the expected monetary value of a decision based on its risk event probability and monetary value

Example



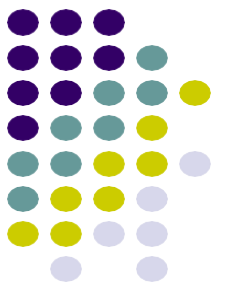
- Project X has a 60% Probability of success with an impact of Birr 50,000 and has a 40% chance of failure with an impact of Birr -20,000. What is the Expected Monetary Value of this Project?

EMV of success = Birr 50,000 * 60% = Birr 30,000

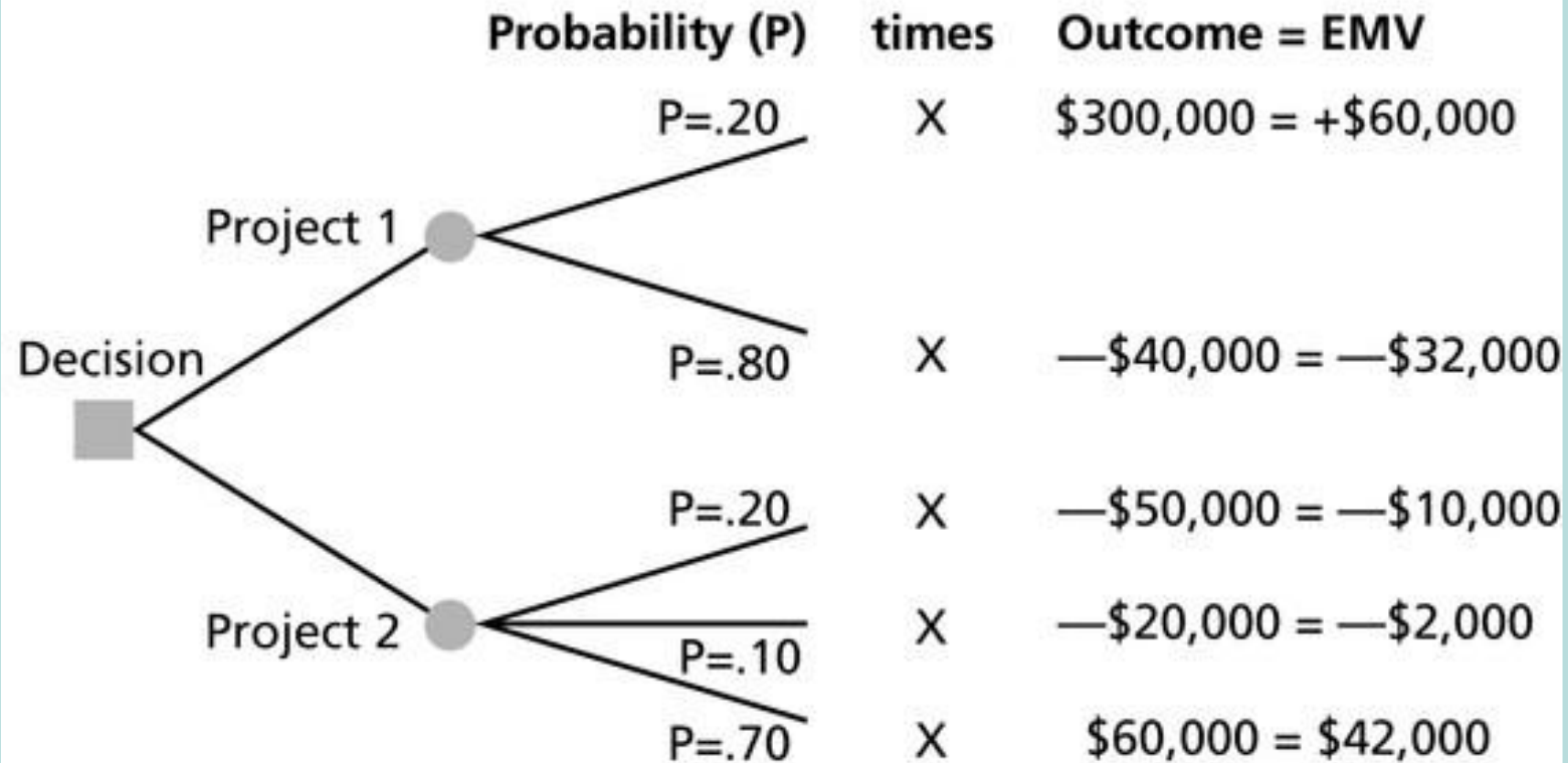
EMV of failure = Birr -20,000 * 40% = Birr -8,000

Total EMV = Birr 22,000

In all cases, a positive EMV indicates that it is an opportunity while a negative EMV indicates a Risk/Threat.



Expected Monetary Value (EMV) Example



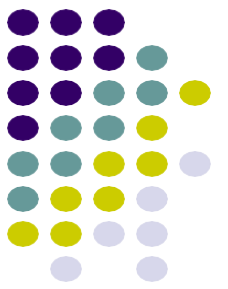
Project 1's EMV = \$60,000 — 32,000 = \$28,000

Project 2's EMV = —\$10,000 — 2,000 + 42,000 = \$30,000

Simulation



- Simulation uses a representation or **model of a system** to **analyze the expected behavior or performance of the system**
- Monte Carlo analysis simulates a model's outcome many times to provide a statistical distribution of the calculated results
- To use a Monte Carlo simulation, you must have three estimates (most likely, pessimistic, and optimistic) plus an estimate of the likelihood of the estimate being between the optimistic and most likely values



Risk Response Planning

- After identifying and quantifying risk, you must decide how to respond to them
- Four main **strategies**:
 - **Risk avoidance**: eliminating a specific threat or risk, usually by eliminating its causes
 - **Risk mitigation**: reducing the impact of a risk event by reducing the probability of its occurrence
 - **Risk transference**: shifting the consequence of a risk and responsibility for its management to a third party
 - **Risk acceptance**: accepting the consequences should a risk occur

Possible Risk Strategies



- Can I **avoid** the risk?
- Can I **reduce** the risk impact or
Can I **reduce** the risk probability?
- Can I **limit** the risk?
(Contingency)?
- Can I **transfer** the risk?
- Can I **accept** the risk ?



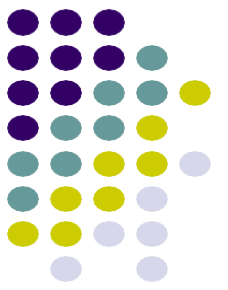
Risk
Reduction
Staircase



GENERAL RISK MITIGATION STRATEGIES FOR TECHNICAL, COST, AND SCHEDULE RISKS

TECHNICAL RISKS	COST RISKS	SCHEDULE RISKS
Emphasize team support and avoid stand-alone project structure	Increase the frequency of project monitoring	Increase the frequency of project monitoring
Increase project manager authority	Use WBS and CPM	Use WBS and CPM
Improve problem handling and communication	Improve communication, project goals understanding, and team support	Select the most experienced project manager
Increase the frequency of project monitoring	Increase project manager authority	
Use WBS and CPM		

Risk Monitoring and Control



- Monitoring risks involves knowing **their status**
- Controlling risks involves carrying out the **risk management plans** as risks occur
- The main outputs of risk monitoring and control are **corrective action, project change requests, and updates to other plans**

Thank You!

